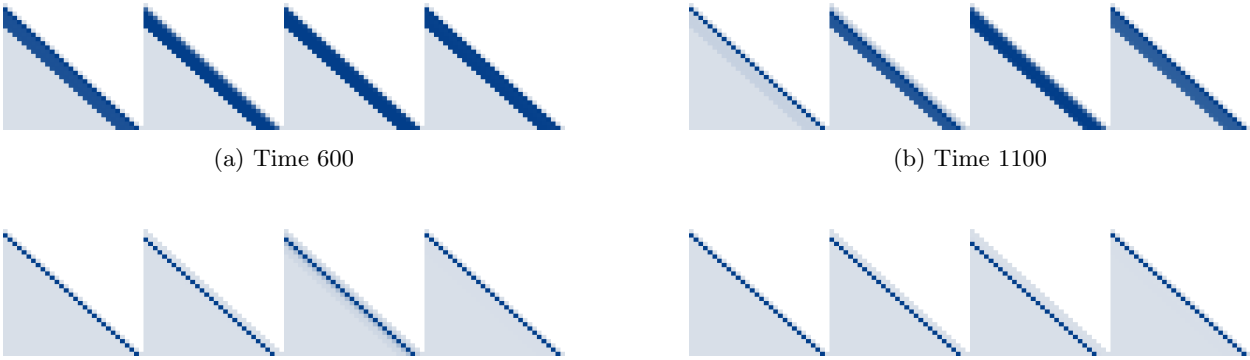
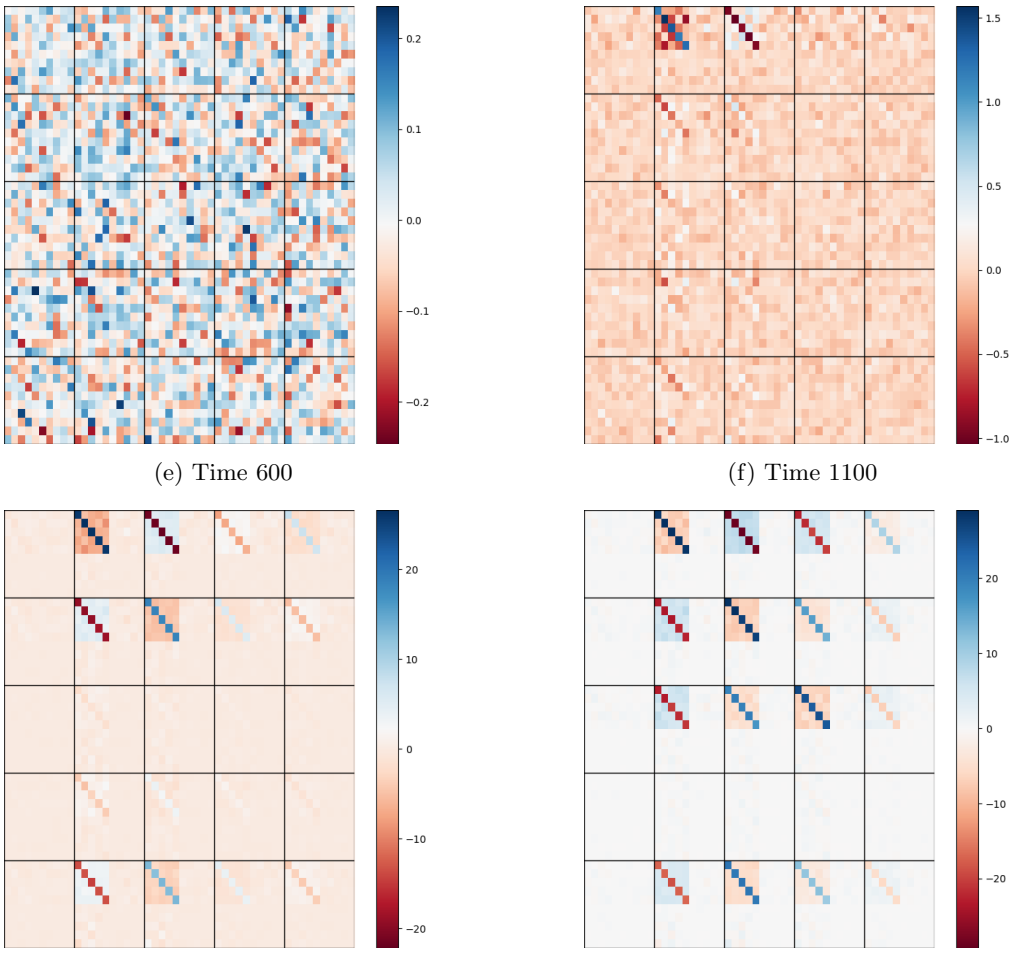
Attention Layer 1



(a) Time 600 (b) Time 1100

(c) Time 1600 (d) Time 2300

$Q^\top K$ Layer 2



(e) Time 600 (f) Time 1100

(g) Time 1600 (h) Time 2300

Figure 1: **Q,K in second layer with random initialization.** Time snapshots of representations of first layer and the second layer for order $k = 5$ and sequence length$l = 120$ without assumption A1, A2. We use Gaussian initialization for Q and K with variance 0.1, making $Q^\top K$ random at initialization as is evident from time 600. We see the evolution of the second layer where the blocks aligns with $\tilde{I}$. Although other blocks also move, the blocks directly above the principal diagonal dominates and the evolution stays close to our simplification.

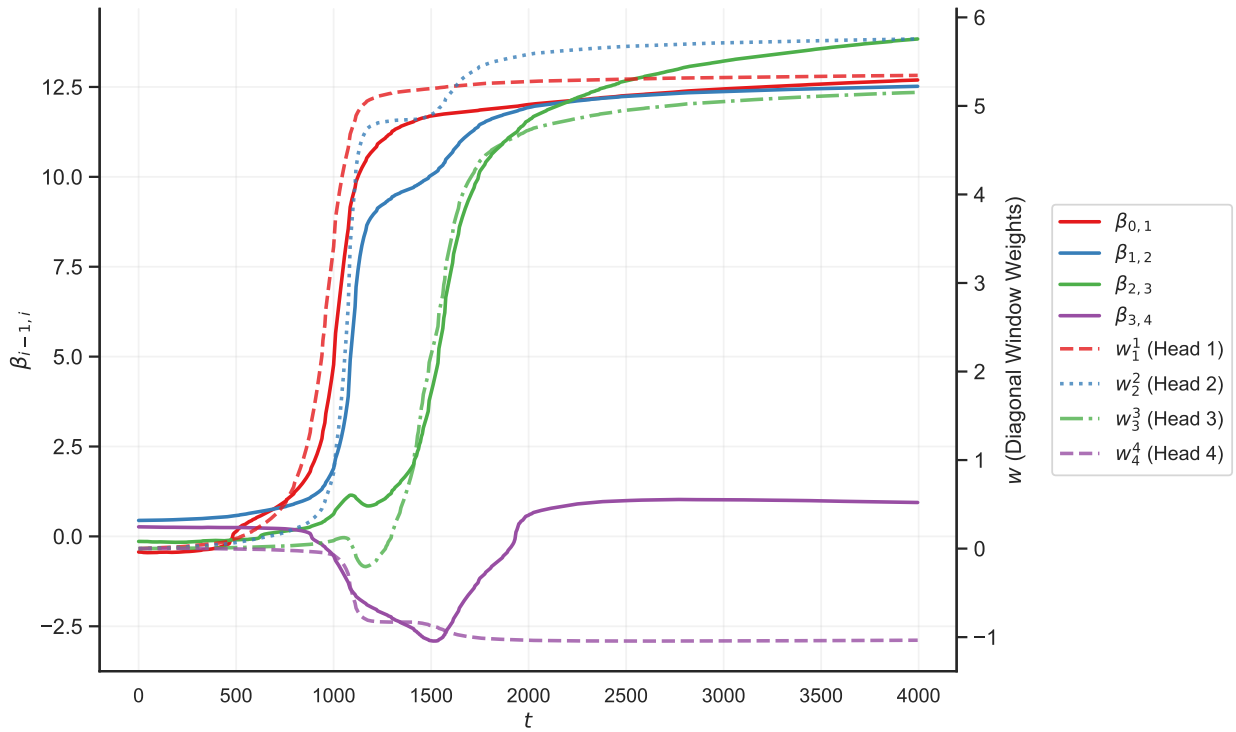


Figure 2: **Betas and Window Weights with full Q,K initialization.** Again for the same setting as Figure 1. We plot the attention weights and also the equivalent β's. We compute the betas as following: $\beta_{h,g}^2 = |\langle B_{h,g}, I\rangle|$, where $B_{h,g}$ is block along row h and column g. We see that the β's, attention weights are tied and evolve in a way that is consistent with our conservative law and theory. The triaing is not complete as it is taking a long time to leave the last plateau.

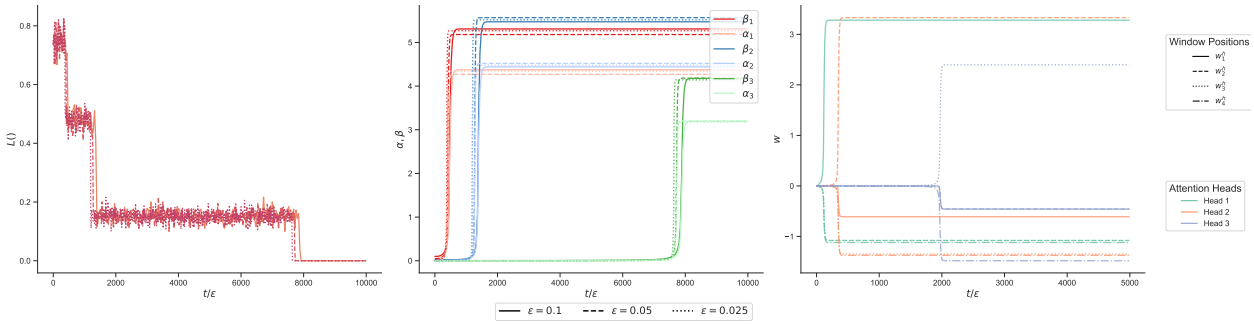


Figure 3: **Experiment with incomplete permutation list in the context.** The order $k = 4$ and sequence length $l = 60$. We plot the evolution of runs with various scales of initialization trained with cross entropy and with assumptions A(0)-A(3). It validates the timescales and conservative law when all the permutations are not present in the context. The remaining experimental setup is same as Figure 11 in the main paper.